

ALAN DA MATTA

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Texas

PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years building and operating backend systems in healthcare and regulated financial environments. At JPMorgan Chase, I have driven a reduction of Kubernetes memory usage by 30% and built a Python library that auto-generates classes mirroring our data store structure, reducing data patching from hours to minutes and catching TDQ errors locally before Spark jobs execute. I gravitate toward the operational gaps nobody's fixed yet and build tooling to close them.

WORK EXPERIENCE

JP Morgan Chase | Software Engineer On-site · Allen, TX · March 2024 - Present

Regulatory Data team — building shared platform solutions for data pipeline automation and supporting product teams across the organization.

- Leading a modernization initiative to migrate legacy **PL/SQL** procedures to **PySpark**, reducing pipeline complexity and improving scalability for the Regulatory Data platform.
- Architected and built an internal **Python** library that auto-generates type-safe classes mirroring datasets from the our data store, establishing a formal contract between the data lake and consuming services.
- Developed agentic AI workflows and reusable skills to automate repetitive migration tasks, accelerating the team's throughput on the modernization effort.
- Mentored and guided 3+ product teams on **Java** and **Spring** best practices, accelerating feature delivery and reducing production defects.
- Drove CI/CD onboarding for multiple product teams through office hours and live training sessions.
- Designed and delivered a proof-of-concept to automate internal data modeling workflows, reducing manual effort from hours to minutes.
- Optimized compute usage on **Kubernetes**, cutting resource consumption by **30%** and reducing infrastructure costs.
- Migrated applications from on-premises infrastructure to the public cloud.
- Maintained availability and reliability of shared services on **Kubernetes**.

Kin + Carta | Technical Consultant Full Remote · April 2022 - March 2024

Embedded consultant for a major US financial institution — focused on customer-facing platform reliability and feature delivery.

- Migrated **Spring Boot** applications from **PCF** to **OCP**, improving platform resilience and deployment flexibility.
- Ensured service availability in a **Kubernetes** environment with strict security and privacy requirements, driving alerting systems adoption and triage automation.
- Developed dashboards and automated reports in **Kibana** for executive visibility into systems health, increasing oversight and confidence on production deployments.
- Improved log readability and distributed tracing, reducing mean time to resolution for production incidents.
- Integrated applications with **Instana**, delivering improved alerting and observability.
- Drove team's documentation output for team onboarding, operational runbooks, and issues resolutions.

Tata Consultancy Services | Java Web Developer Londrina, Brazil · July 2018 - May 2021

Consultant for Invisalign's physician portal — enabling doctors to create and manage patient orthodontic treatment plans.

- Designed and implemented a configuration comparison tool for orthodontic treatment planning environments, improving accuracy of environment parity checks.
- Built a **React.js** application for portal configuration management.
- Developed and maintained containerized services in **Spring Boot**; supported legacy applications in **Spring MVC** and **Struts 2**.
- Wrote unit and integration tests in **JUnit** and **Mockito**, ensuring solution correctness and regression coverage.
- Developed and optimized SQL queries in **PostgreSQL**.
- Integrated internal systems with third-party services.

Vitai Solutions | Java Web Developer Apucarana, Brazil · August 2017 - July 2018

Web development for a hospital management system built on a Java monolith.

- Proposed and built a functional demo of migration of the patient creation module from legacy Struts 1 to **Angular 2**.
- Implemented features and resolved defects in a monolithic **Java/Struts 1** application.
- Optimized complex SQL queries for clinical data operations, reducing patient intake searches measurably faster.

PROJECTS

Plum | Python Superset Language with Native AI Syntax plumlang.dev · github.com/becarder/plumlang

- Designed and built Plum, a transpiler that extends Python with a native AI operator — **?[prompt | model]** — enabling inline LLM calls as first-class expressions, compiled to valid Python.
- Reduces multi-line LLM boilerplate to a single expression, making AI-powered scripting, data pipelines, and prototyping accessible without wiring up full API clients.
- Published on PyPI (**pip install plumlang**) and available at plumlang.dev with full documentation and examples.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, PL/SQL

Data Engineering: PySpark, Apache Spark, data pipeline automation

Frameworks: Spring Boot, Spring MVC, React.js, Next.js

Cloud & DevOps: Kubernetes, Docker, AWS, PCF, OCP, Harness.io, Jenkins

Databases: PostgreSQL, Oracle

Observability: Instana, Kibana, Splunk

EDUCATION

Specialization in Software Engineering and Databases Londrina, Brazil · December 2019

State University of Londrina

Bachelor of Information Systems Apucarana, Brazil · December 2016

Apucarana College